

Image Morphology

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Introduction

- “***Morphology*** “ – a branch in biology that deals with the form and structure of animals and plants.
- “***Mathematical Morphology***” – as a tool for extracting image components, that are useful in the representation and description of region shape.
- The language of mathematical morphology is – **Set theory**.
- Unified and powerful approach to numerous image processing problems.
- In binary images , the set elements are members of the **2-D** integer space – $\mathbb{Z}^2$. where each element (x,y) is a coordinate of a black (or white) pixel in the image.

Set Theory

Basic Concepts in Set Theory

Subset

$$A \subseteq B$$

Union

$$A \cup B$$

Intersection

$$A \cap B$$

disjoint / mutually exclusive

$$A \cap B = \emptyset$$

Complement

$$A^c \equiv \{w \mid w \notin A\}$$

Difference

$$A - B \equiv \{w \mid w \in A, w \notin B\} = A \cap B^c$$

Reflection

$$\mathcal{B} \equiv \{w \mid w = -b, \quad \forall b \in B\}$$

Translation

$$(A)z \equiv \{c \mid c = a + z, \quad \forall a \in A\}$$

Logic Operations

The principal logic operations used in image processing
AND, OR, NOT (COMPLEMENT).

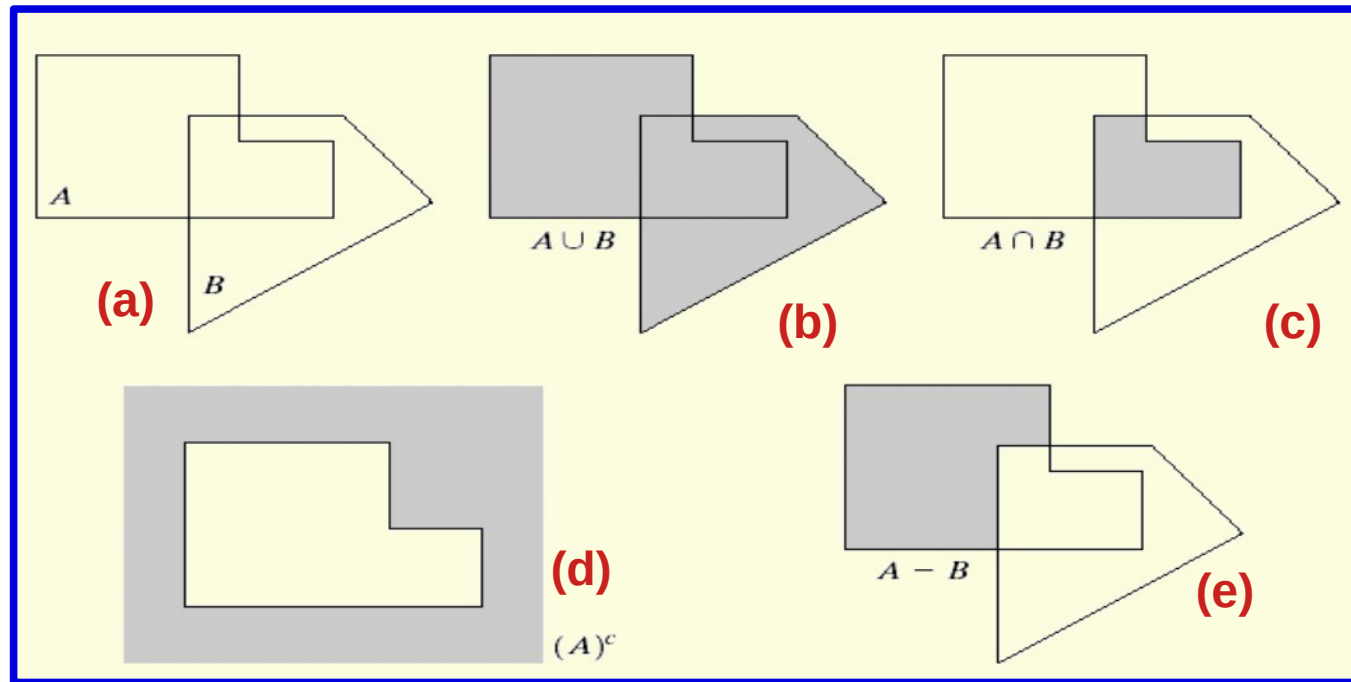
- Logic operations are preformed on a pixel by pixel basis between corresponding pixels (bitwise).

Other important logic operations :
XOR (exclusive OR), NAND (NOT-AND)

Logic operations are just a private case for a **binary set operations**, such :

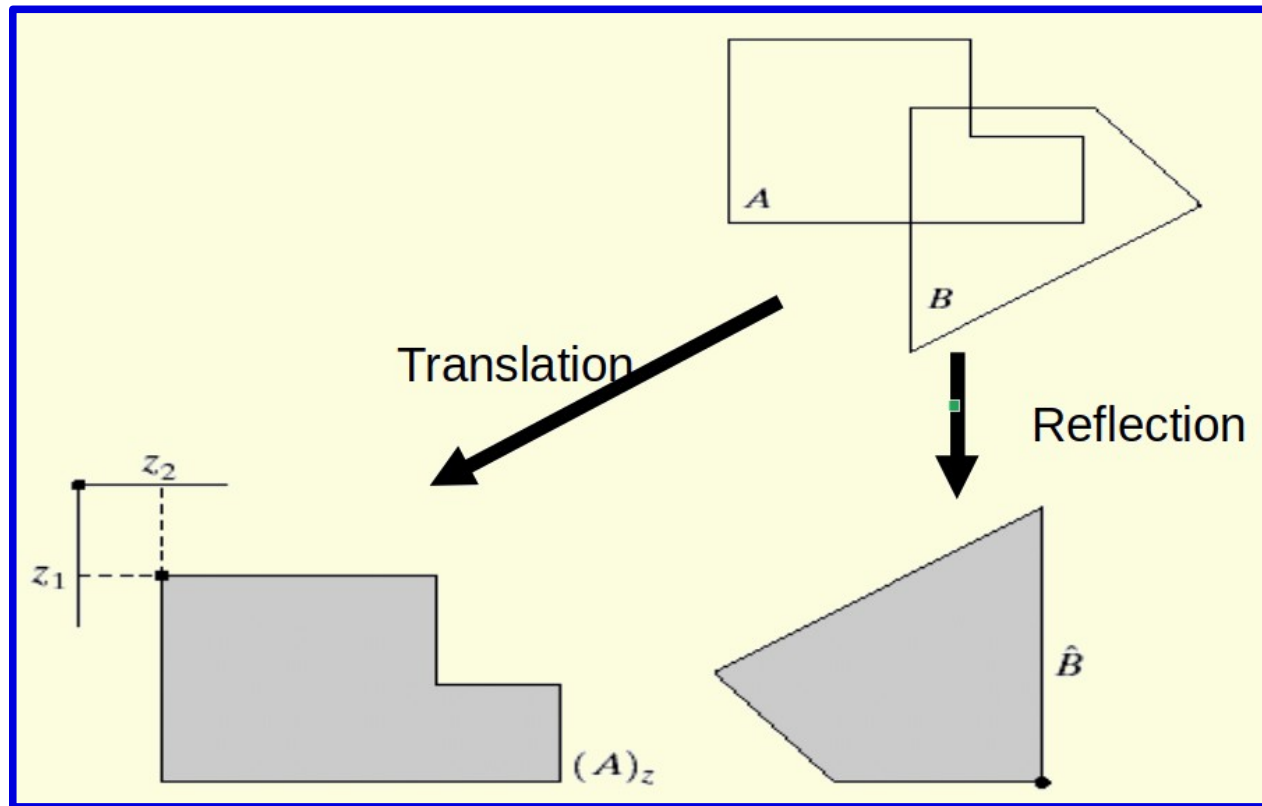
AND – Intersection , **OR** – Union, **NOT**-Complement.

Some Basic Concepts from Set Theory



- (a) Two Sets A and B
- (b) Union of A and B
- (c) Intersection of A and B
- (d) Complement of A
- (e) Difference between A and B

Some Basic Concepts from Set Theory

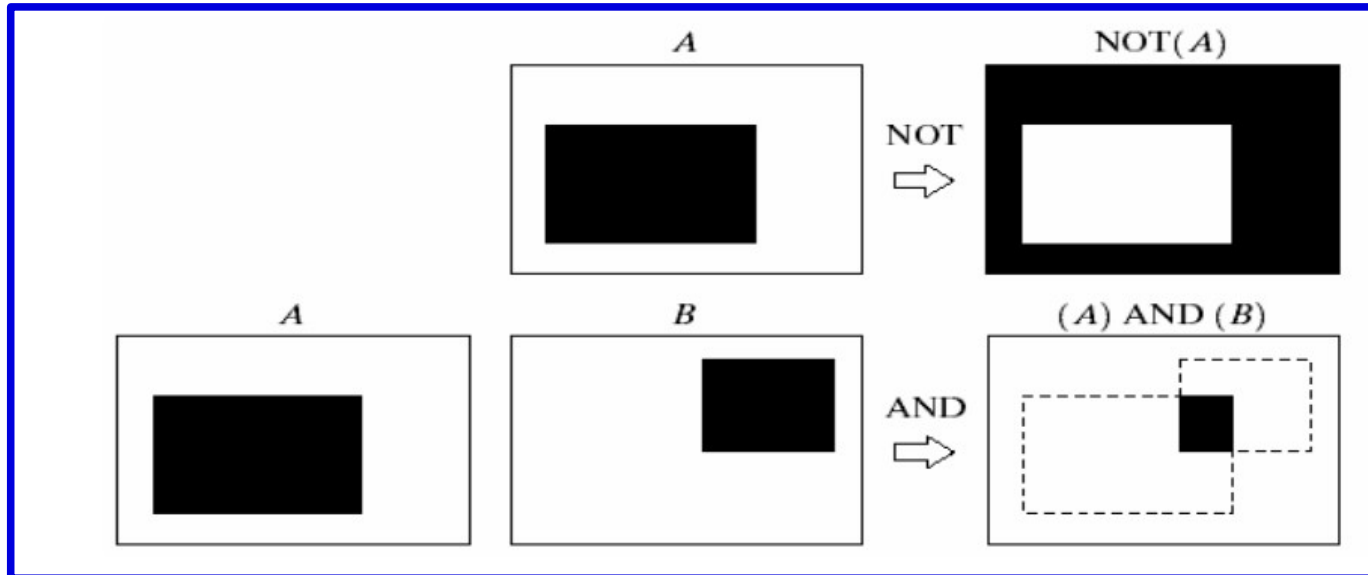


- (a) Translation of A by Z
(b) Reflection of B

Logic Operations Involving Binary Images

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p	q	$p \text{ AND } q \text{ (also } p \cdot q \text{)}$	$p \text{ OR } q \text{ (also } p + q \text{)}$	$\text{NOT } (p) \text{ (also } \bar{p} \text{)}$
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

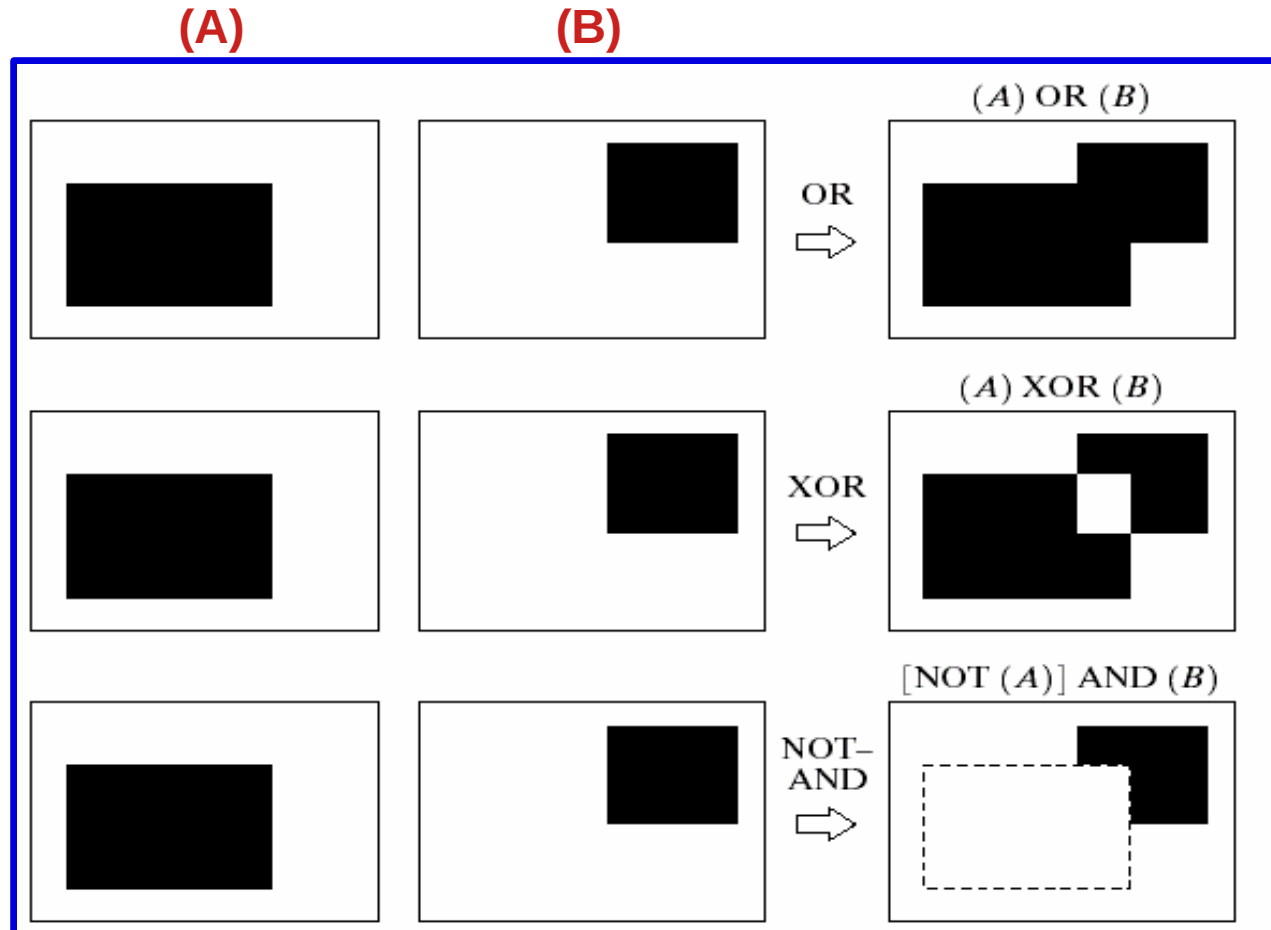


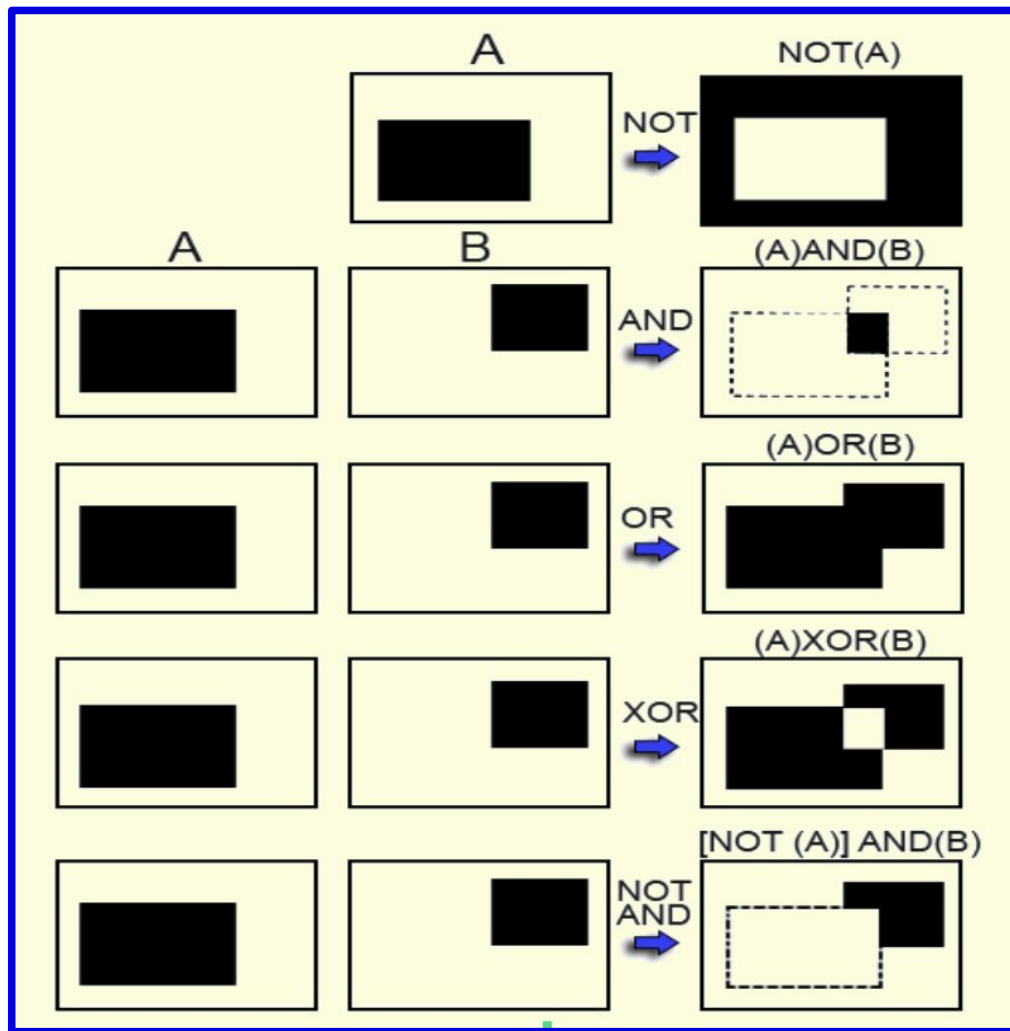
Black Represents Binary “1”

White Represents Binary “0”

Logic Operations Involving Binary Images

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Introduction to Morphological Operators

- **Morphological Operators**
 - Used generally on binary images, e.g., background subtraction results!
 - Used on gray value images, if viewed as a stack to binary images.
- Good for, e.g.,
 - Noise removal in background
 - Removal of holes in foreground / background

Dilation & Erosion

- Basic operations
- Are dual to each other:
 - Erosion shrinks foreground, enlarges Background
 - Dilation enlarges foreground, shrinks background

[1] Dilation

- **Dilation** is the set of all points in the image, where the structuring element “touches” the foreground.
- Consider each pixel in the input image
 - If the structuring element touches the foreground image, write a “1” at the origin of the structuring element!
- **Input:**
 - **Binary Image**
 - **Structuring Element, containing only 1s!!**

Dilation

- With A and B as sets in Z^2 , the dilation of A by B is defined as

$$A \oplus B = \left\{ z \mid (\hat{B})_z \cap A \neq \varnothing \right\}$$

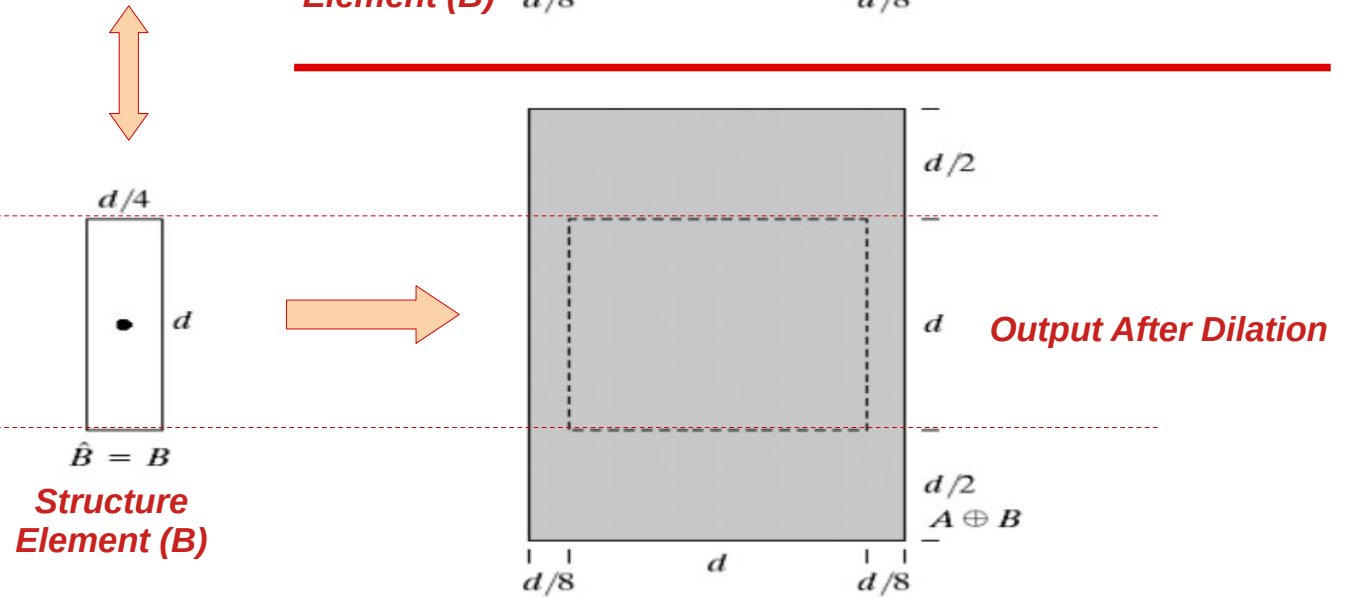
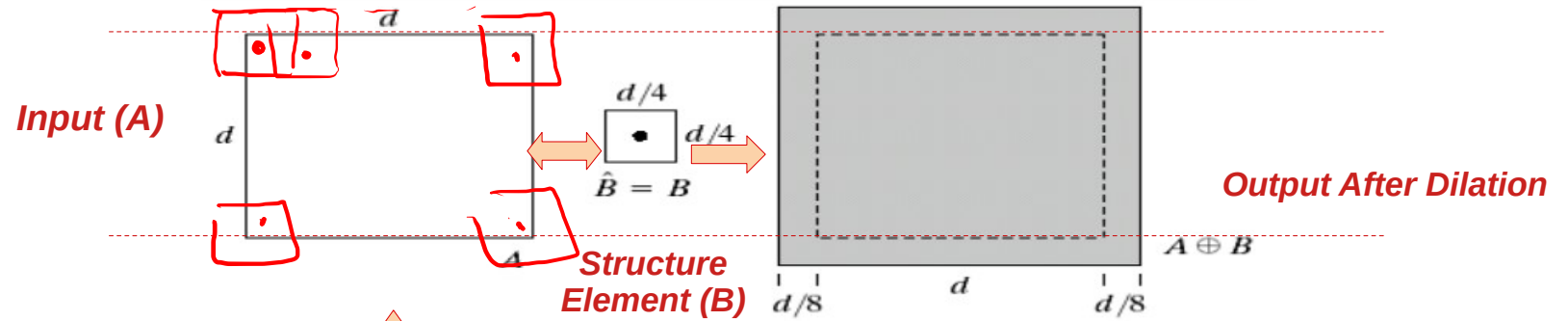
where $\hat{B}$: the reflection of B about its origin and shifting this reflection by z

- The dilation of A by B is the set of all displacements, z , such that $\hat{B}$ and A **overlap by at least one element**. Thus,

$$A \oplus B = \left\{ z \mid [(\hat{B})_z \cap A] \subseteq A \right\}$$

- Set B is referred to as the **structuring element** in dilation.

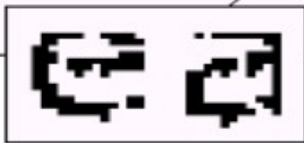
Dilation



Dilation - Example

(a)

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



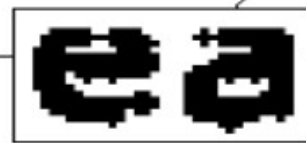
*Sample Text,
Poor Resolution,
Broken characters*

(b)

0	1	0
1	1	1
0	1	0

**Structuring
element**

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

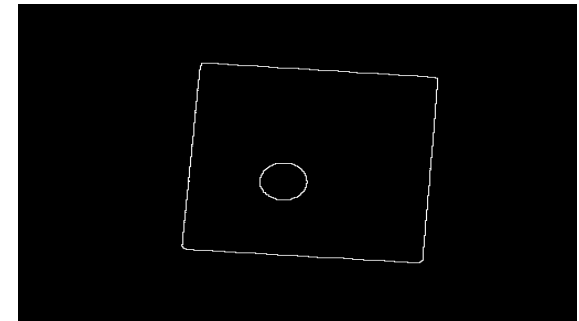
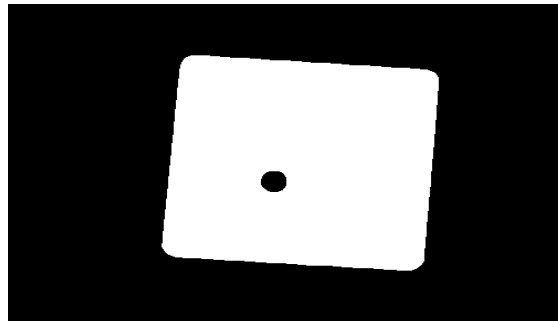
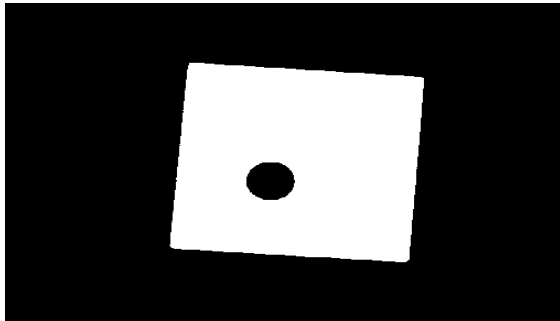


*Dilation of (a) by (b),
Broken segments are joined*

Edge Detection

Edge Detection

1. Dilate input image (White Pixels)
2. Subtract input image from dilated image
3. Edges remain!

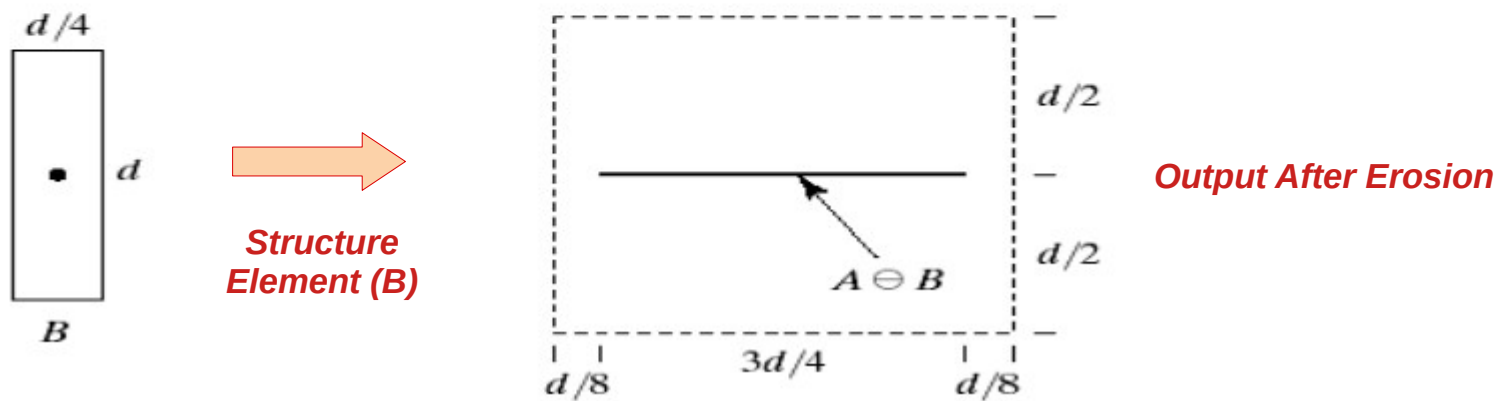
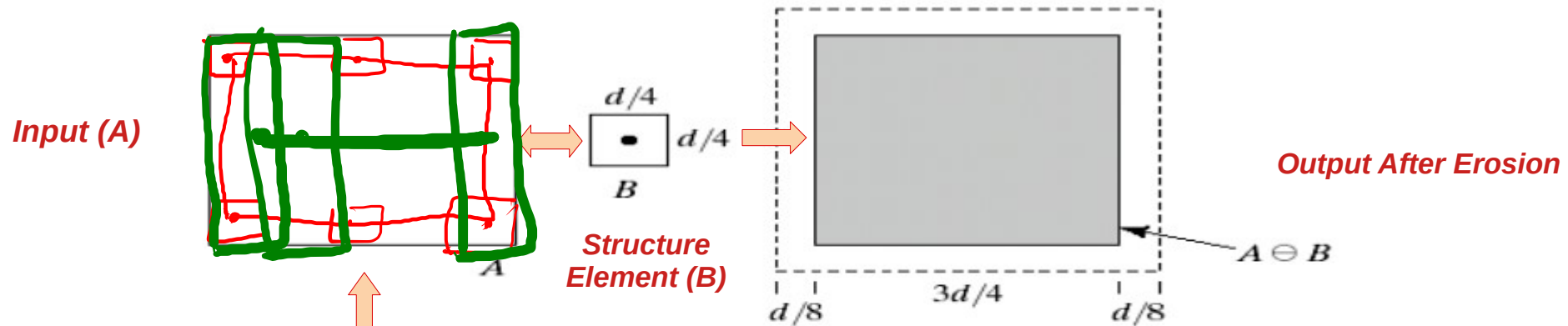


[2] Erosion

Erosion

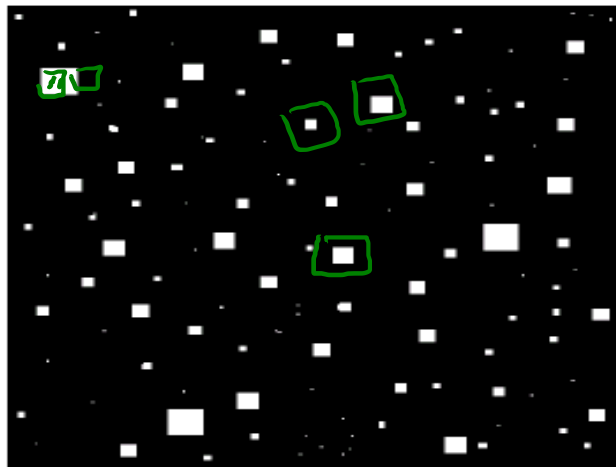
- **Erosion** is the set of all points in the image, where the structuring element **“fits into”**.
- Consider each foreground pixel in the input image
 - If the structuring element fits in, write a “1” at the origin of the structuring element!
- Simple application of **pattern matching**
- **Input:**
 - **Binary Image**
 - **Structuring Element, containing only 1s!**

Erosion

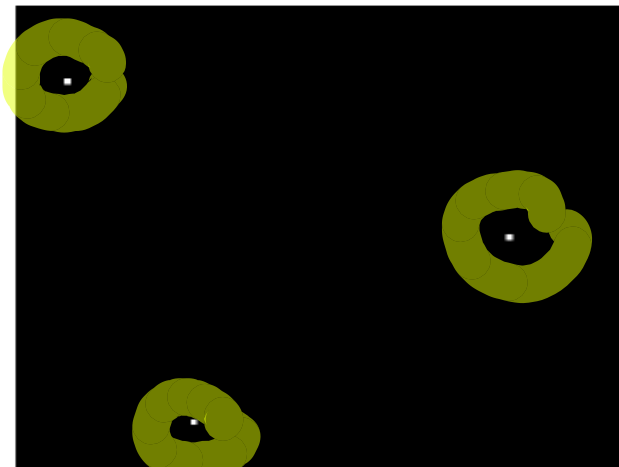


Erosion Then Dilation Example

☐ Squares of Different Sizes
(1,3,5,7,9,15)



Erosion with square of size 13
(All ones)

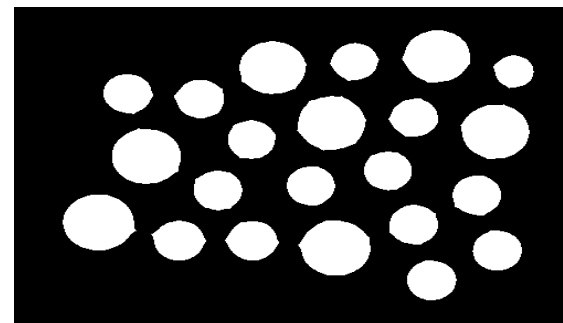
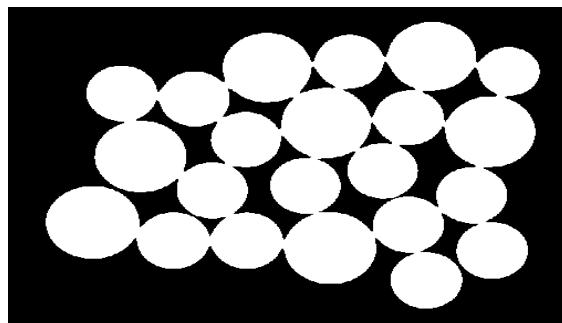
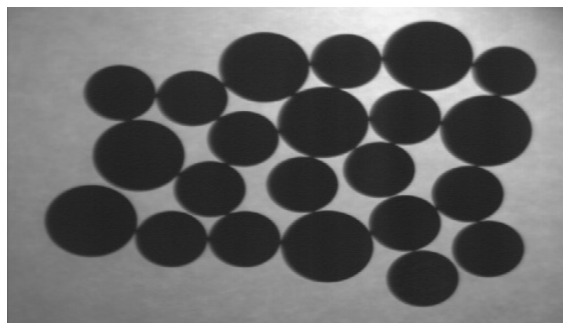


Dilation with square of size 13
(All ones)



Counting Coins

- Counting coins is difficult because they touch each other!
- Solution: Binarization and Erosion separates them!



Duality between dilation and erosion

- Dilation and erosion are duals of each other with respect to set complementation and reflection. That is,

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

- One of the simplest uses of erosion is for eliminating irrelevant details (in terms of size) from a binary image.

Opening and closing

Opening & Closing

- Important operations
- Derived from the fundamental operations
 - Dilatation
 - Erosion
- Usually applied to binary images, but gray value images are also possible
- Opening and closing are dual operations

Opening & Closing

- Opening:

- smooth the contour of an object, break narrow isthmuses, and eliminate thin protrusions.
- The opening A by B is the erosion of A by B , followed by a dilation of the result by B

$$A \circ B = (A \ominus B) \oplus B$$

- Closing:

- smooth sections of contours but it generally fuses narrow breaks and long thin gulfs, eliminates small holes, and fills gaps in the contour.

$$A \cdot B = (A \oplus B) \ominus B$$

Opening & Closing

- Opening – smooths contours , eliminates protrusions
- Closing – smooths sections of contours, fuses narrow breaks and long thin gulfs, eliminates small holes and fills gaps in contours
- These operations are dual to each other
- These operations can be applied few times, but has effect only once

[3] Opening

Opening

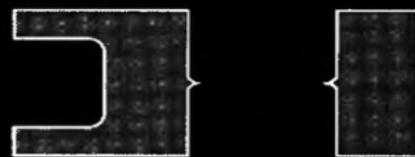
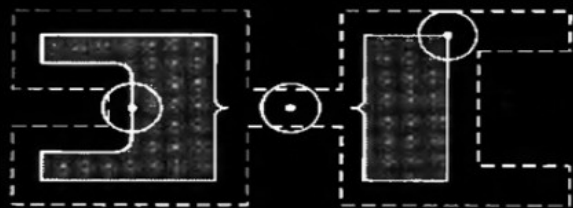
- Take the structuring element (SE) and slide it around *inside* each foreground region.
 - All pixels which can be covered by the SE with the SE being entirely within the foreground region will be preserved.
 - All foreground pixels which can *not* be reached by the structuring element without lapping over the edge of the foreground object will be eroded away!
- Opening is **idempotent**: Repeated application has no further effects!

- Opening –
 - First – erode A by B , and then dilate the result by B
 - In other words, opening is the unification of all B objects Entirely Contained in A

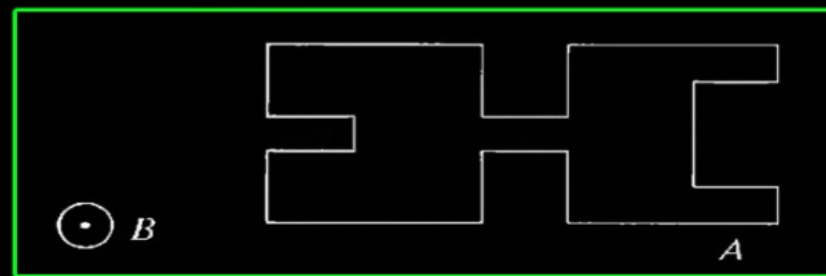
$$A \circ B = (A \ominus B) \oplus B$$

opening

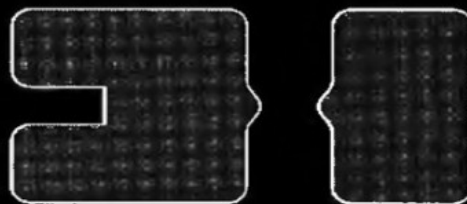
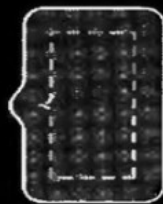
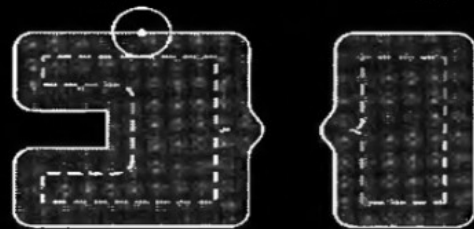
Erosion



$$A \ominus B$$



Opening



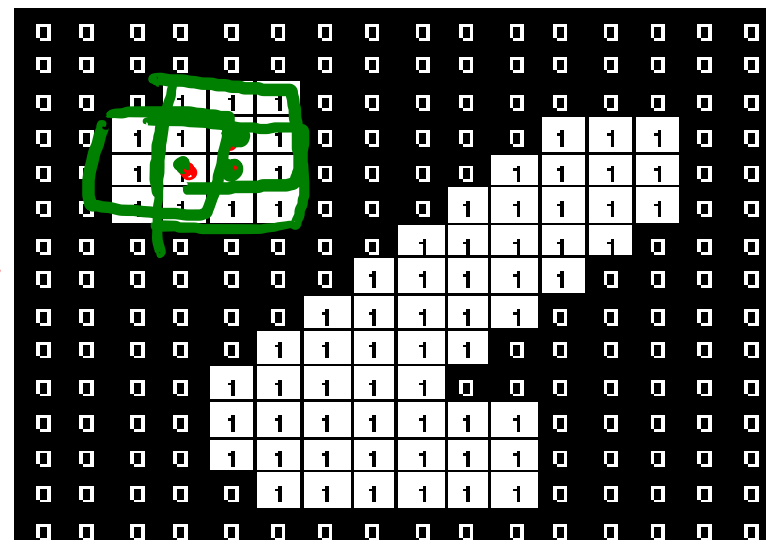
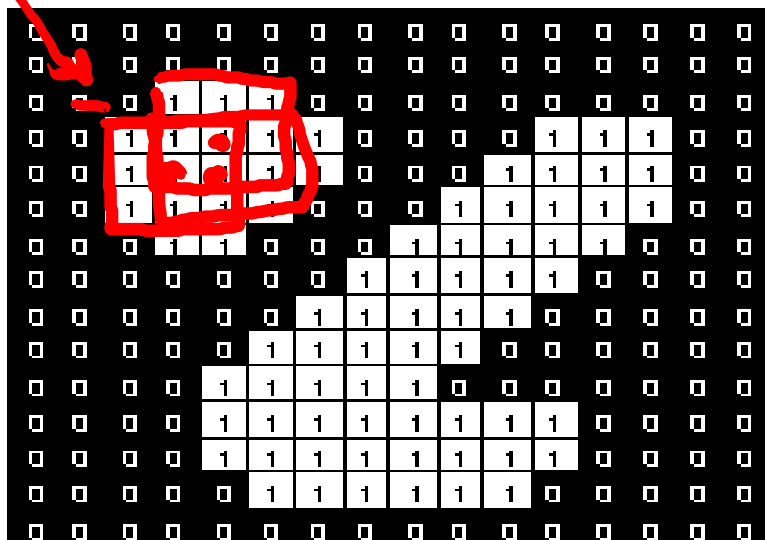
$$A \circ B = (A \ominus B) \oplus B$$

Dilation

Opening

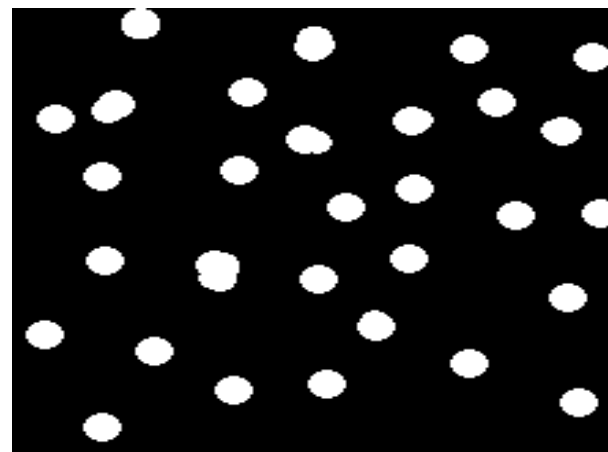
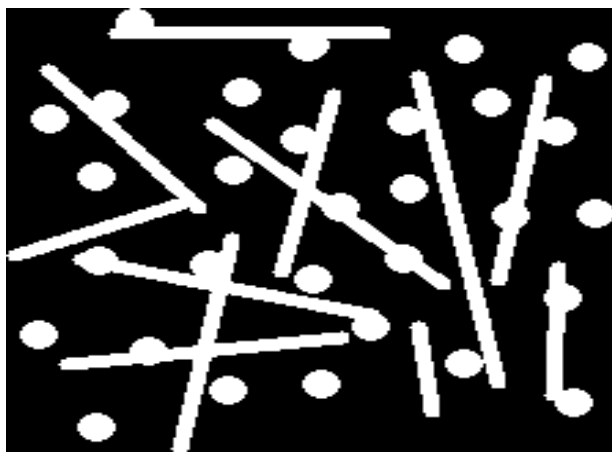
Example

- Structuring element: 3x3 square



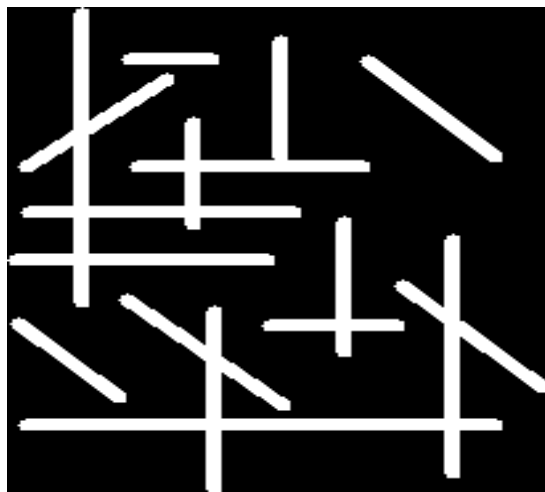
Opening - Example

- Opening with a 11 pixel diameter disc

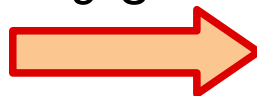


Opening - Example

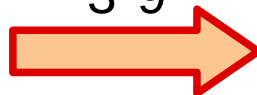
- 3x9 and 9x3 Structuring Element



9*3

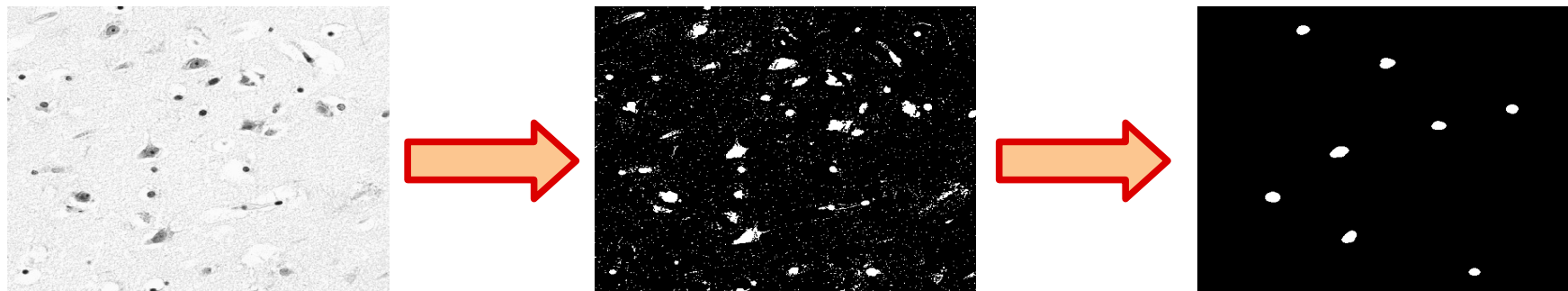


3*9



Use Opening for Separating Blobs

- Use large structuring element that fits into the big blobs
- Structuring Element: 11 pixel disc



[4] Closing

Closing

Closing –

First – dilate A by B , and then erode the result by B

In other words, closing is the group of points, which the intersection of object B around them with object A – is not empty

$$A \cdot B = (A \oplus B) \ominus B$$

Closing

- Take the structuring element (SE) and slide it around *outside* each foreground region.
 - All background pixels which can be covered by the SE with the SE being entirely within the background region will be preserved.
 - All background pixels which can *not* be reached by the structuring element without lapping over the edge of the foreground object will be turned into a foreground.
- Opening is **idempotent**: Repeated application has no further effects!

clustering

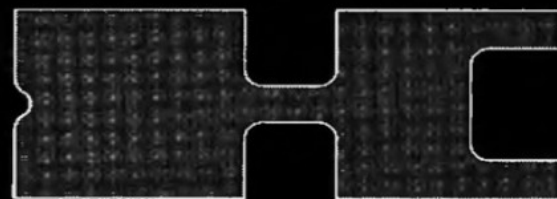
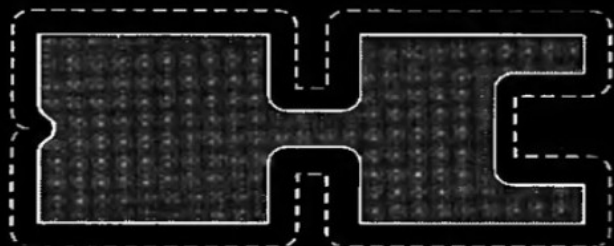
Dilation



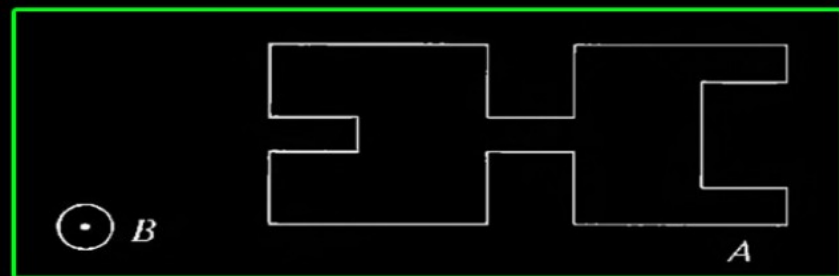
$$A \oplus B$$

Closing

erosion

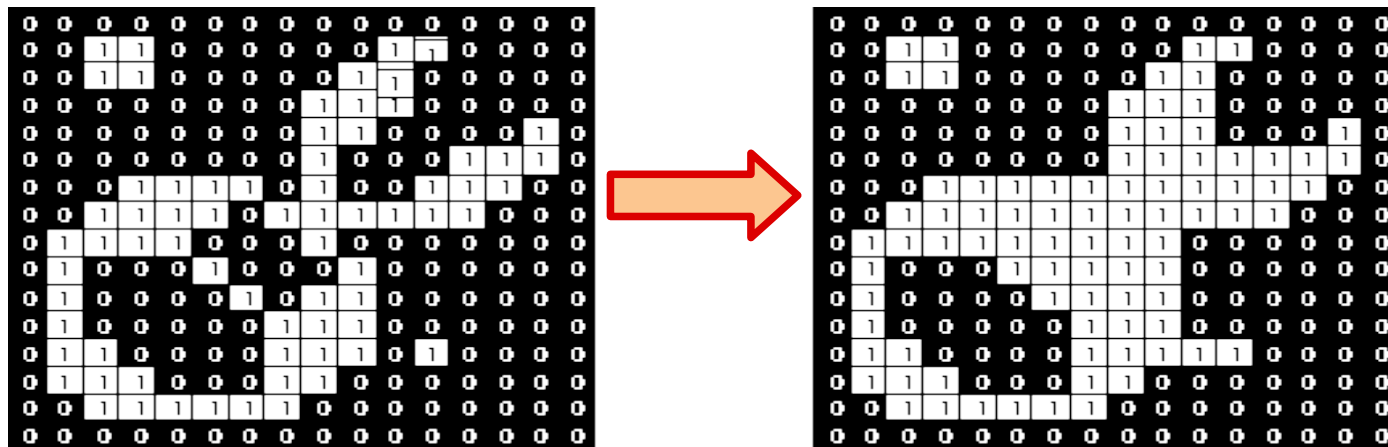


$$A \cdot B = (A \oplus B) \ominus B$$



Closing

- Structuring element: 3x3 square

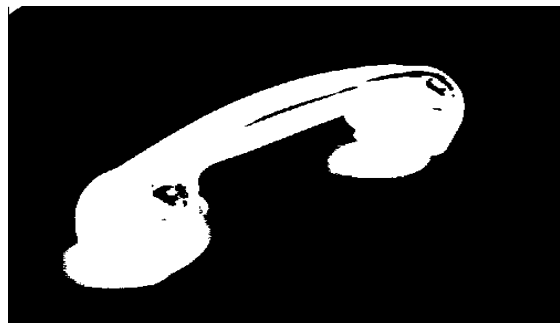


Closing - Example

1. Threshold (Binarization)
2. Closing with disc of size 20



***Input Image (With
Background)***



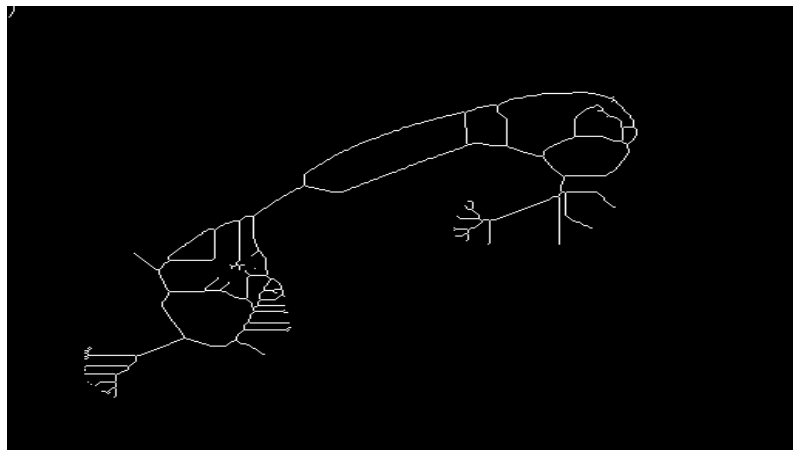
***Image After
Threshold***



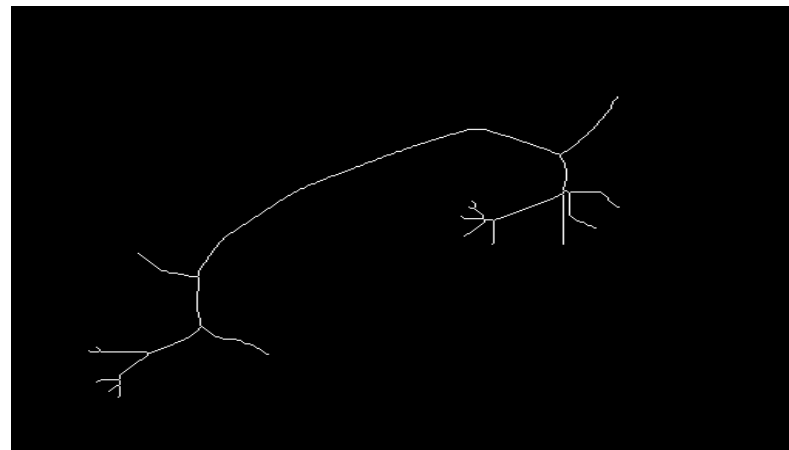
***Image After
Closing***

Closing - Example

- Good for further processing: E.g. Skeleton operation looks better for closed image!



***skeleton of
Thresholded Image***



***skeleton after
Threshold and **close*****

Opening & Closing

- Opening is the *dual* of closing
- *i.e.* opening the foreground pixels with a particular structuring element
- is equivalent to closing the background pixels with the same element.

[5] Hit and Miss

The Hit-and-Miss Transformation

A basic morphological tool for shape detection.

Let the origin of each shape be located at its center of gravity.

If we want to find the location of a **shape**, **X**, at (larger) **image**, **A** :

Let **X** be **enclosed by a small window**, **W**.

The **local background** of **X** with respect to **W** is defined as the *set difference* ($W - X$).

Apply *erosion* operator of **A** by **X**, will get us the set of locations of the origin of **X**, such that **X** is completely contained in **A**.

It may be also view geometrically as the set of all locations of the origin of **X** at which **X** found a match (hit) in **A**.

The Hit-and-Miss Transformation

Apply *erosion* operator on the *complement of A* by the *local background set* ($W - X$).

Notice, that the set of locations for which X **exactly** fits inside A is the ***intersection*** of these two last operators above.

This intersection is precisely the location sought.

Formally:

If B denotes the set composed of X and its background –

$B = (B_1, B_2)$; $B_1 = X$, $B_2 = (W-X)$.

The match (or set of matches) of B in A, denoted is:

$$A \circledast B = (A \ominus B_1) \cap (A^c \ominus B_2)$$

The Hit-and-Miss Transformation

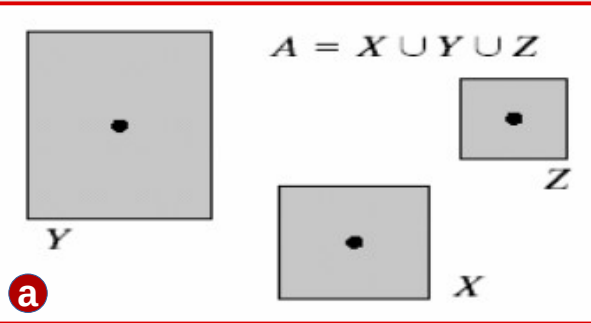
The reason for using these kind of structuring element $B = (B_1, B_2)$ is based on an assumed definition that,

two or more objects are distinct only if they are disjoint (disconnected) sets.

This simplified pattern detection scheme is used in some of the algorithms for – **identifying characters within a text.**

The Hit-and-Miss Transformation (Objective)

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Given:

Set **A** contains three objects **X**, **Y**, **Z**

Objective:

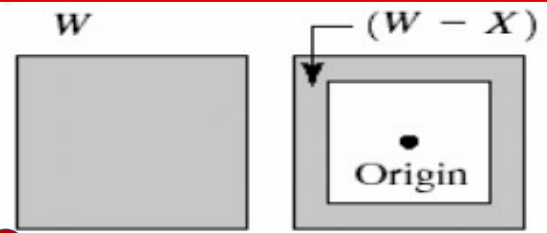
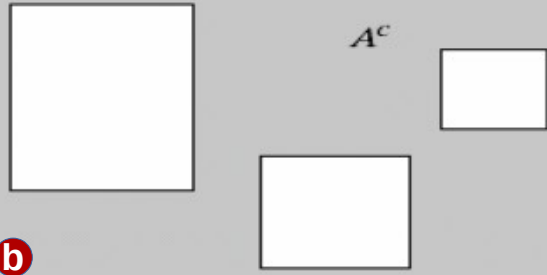
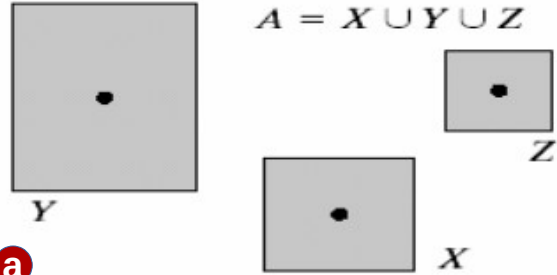
Search for locations of Object **X**

Surrounded by predefined border size
within Set **A**

(a) Set **A**

The Hit-and-Miss Transformation (Inputs)

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Given:

Set A contains three objects **X, Y, Z**

Objective:

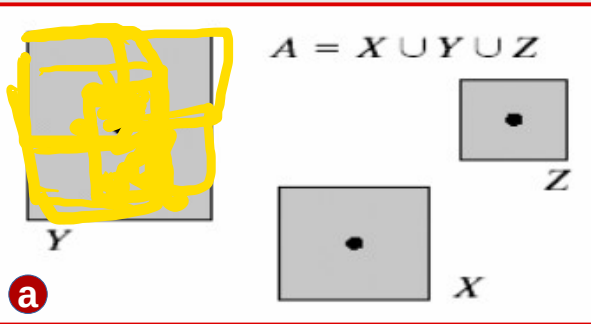
Search for locations of Object **X**

Surrounded by predefined border size
within Set **A**

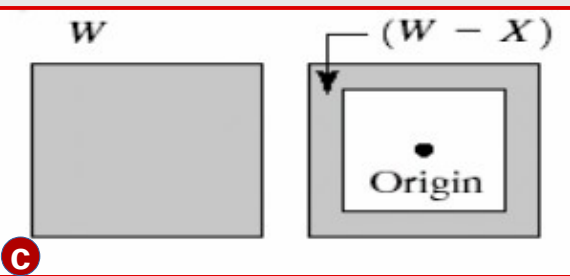
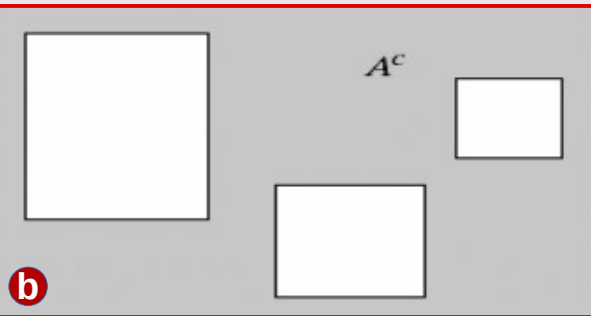
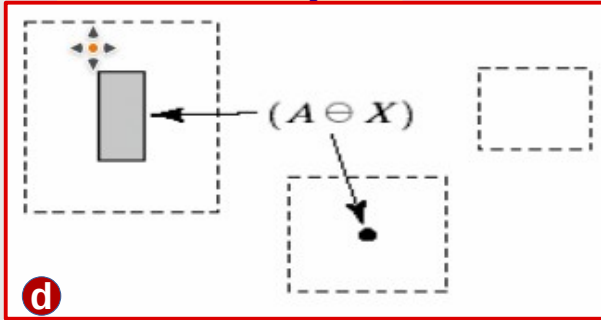
- (a)** Set **A**
- (b)** Complement of **A**
- (c)** Window **W** (larger than **X**)

The Hit-and-Miss Transformation (Step [1])

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Erosion of A by X



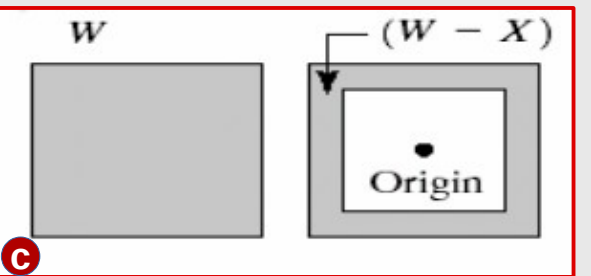
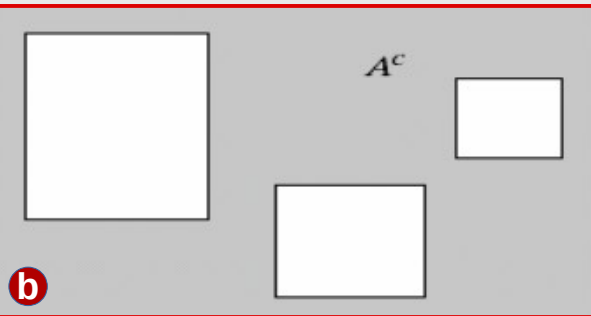
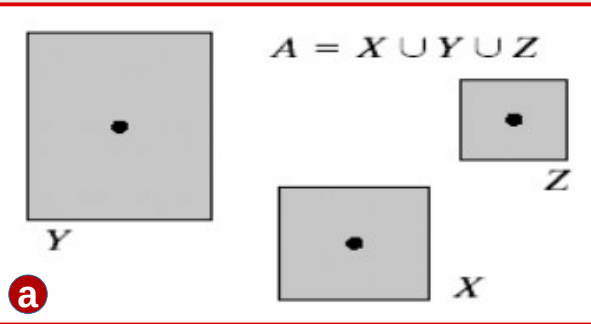
- (a) Set A
- (b) Complement of A
- (c) Window W (larger than X)
- (d) Erosion of A by X

Search for locations of Object X (Surrounded by predefined border) within Set A

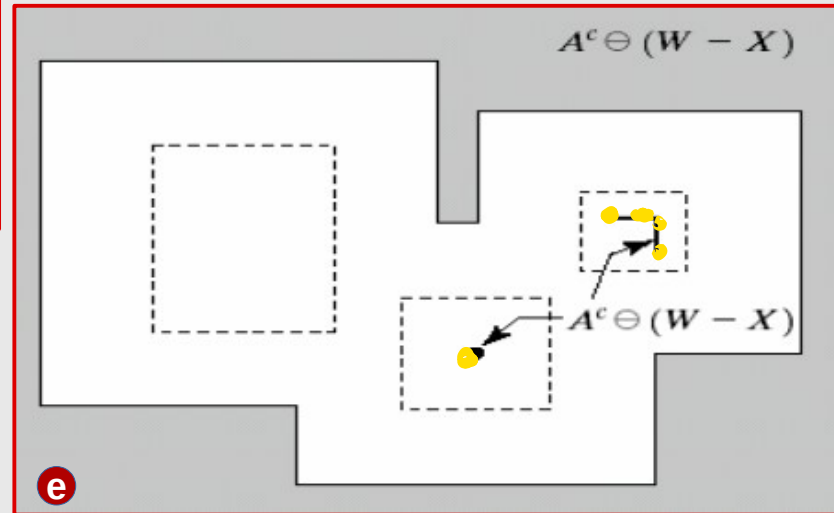
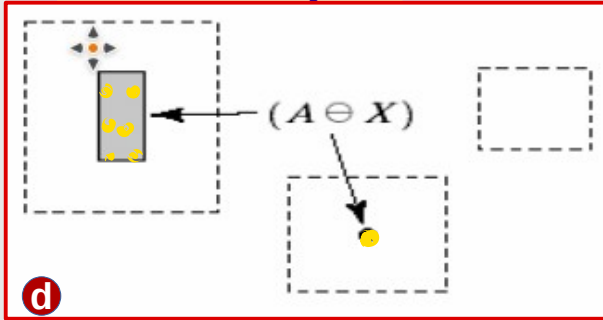
The Hit-and-Miss Transformation

(Step [2])

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Erosion of A by X



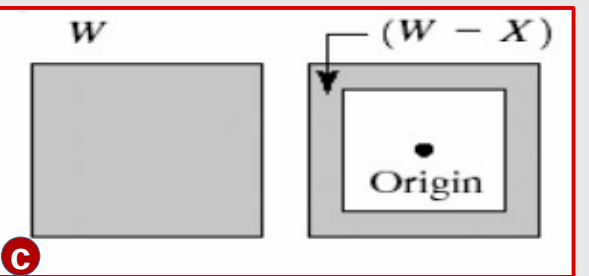
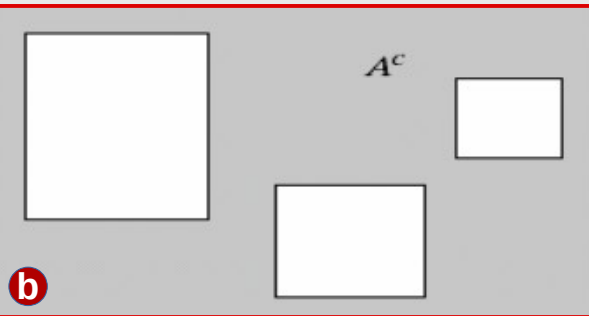
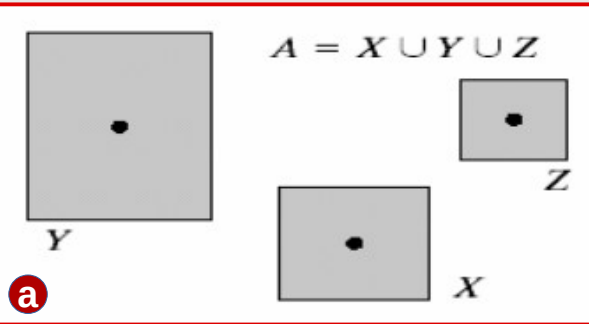
- (a)** Set A
- (b)** Complement of A
- (c)** Window W (larger than X)
- (d)** Erosion of A by X
- (e)** Erosion of A^c by $(W - X)$

Search for locations of Object X (Surrounded by predefined border) within Set A

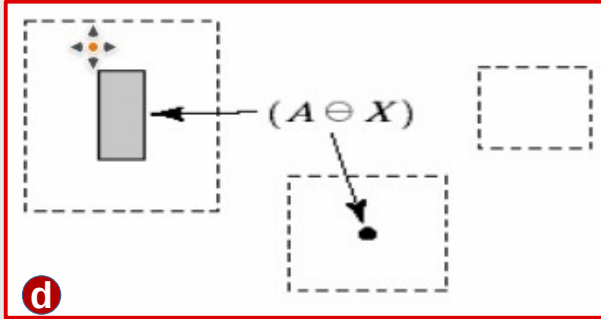
The Hit-and-Miss Transformation

(Step [2])

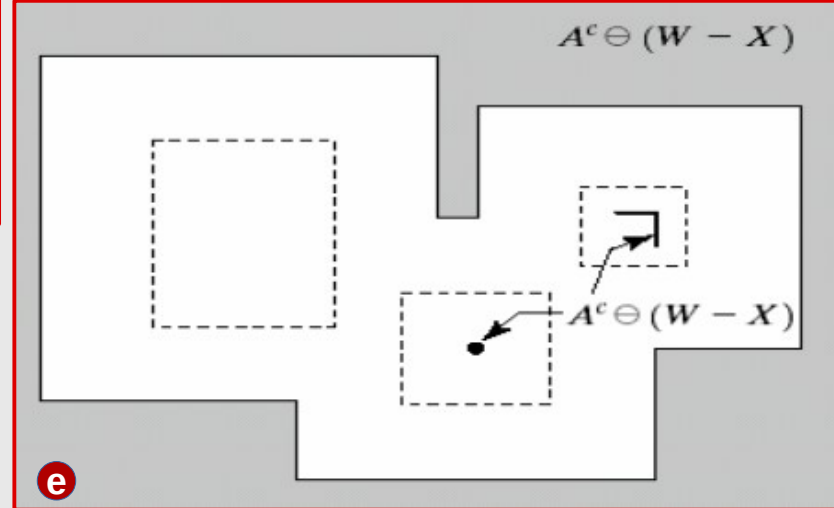
54



Erosion of A by X



Erosion of A^c by $(W - X)$



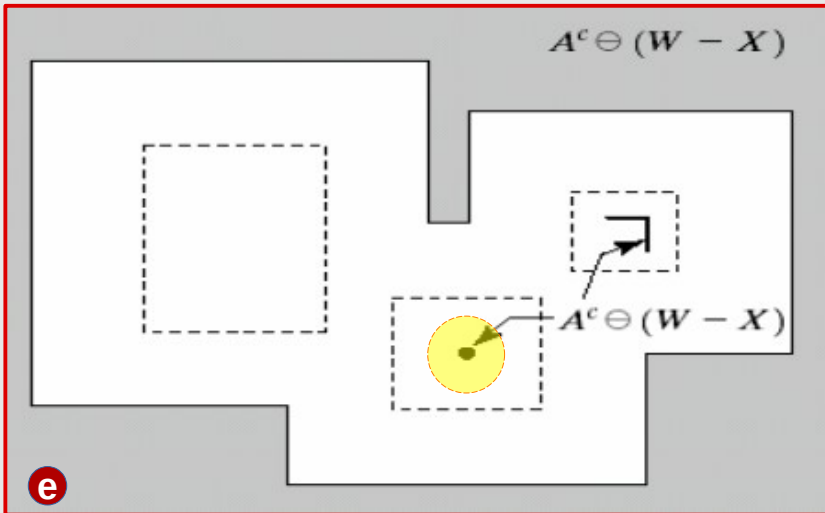
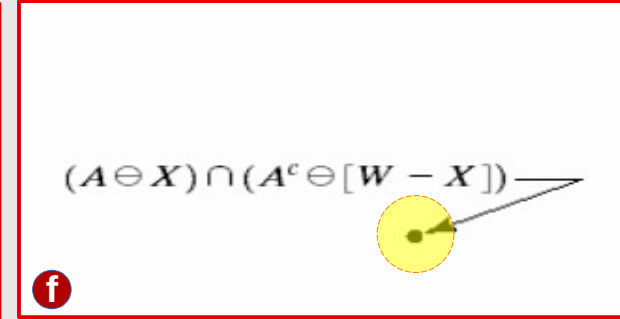
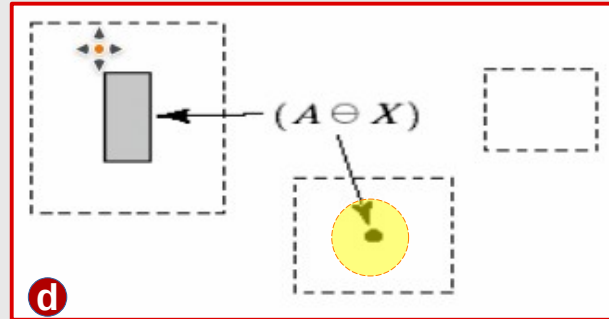
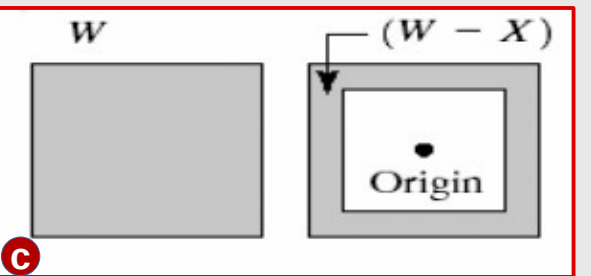
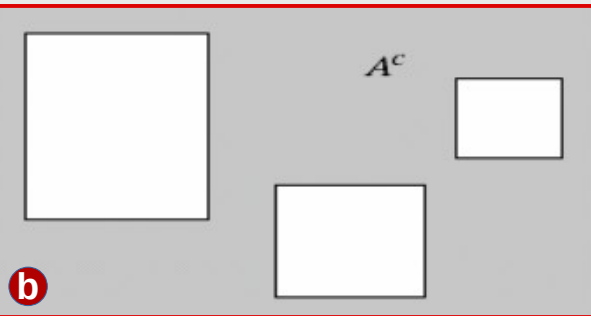
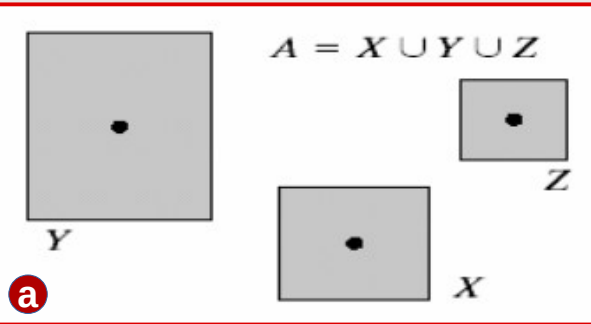
- (a)** Set A
- (b)** Complement of A
- (c)** Window W (larger than X)
- (d)** Erosion of A by X
- (e)** Erosion of A^c by $(W - X)$

Search for locations of Object X (Surrounded by predefined border) within Set A

The Hit-and-Miss Transformation

(Step [3])

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- (a)** Set A
- (b)** Complement of A
- (c)** Window W (larger than X)
- (d)** Erosion of A by X
- (e)** Erosion of A^c by $(W - X)$
- (f)** Intersection of **(d)** and **(e)**

Search for locations of Object X (Surrounded by predefined border) within Set A

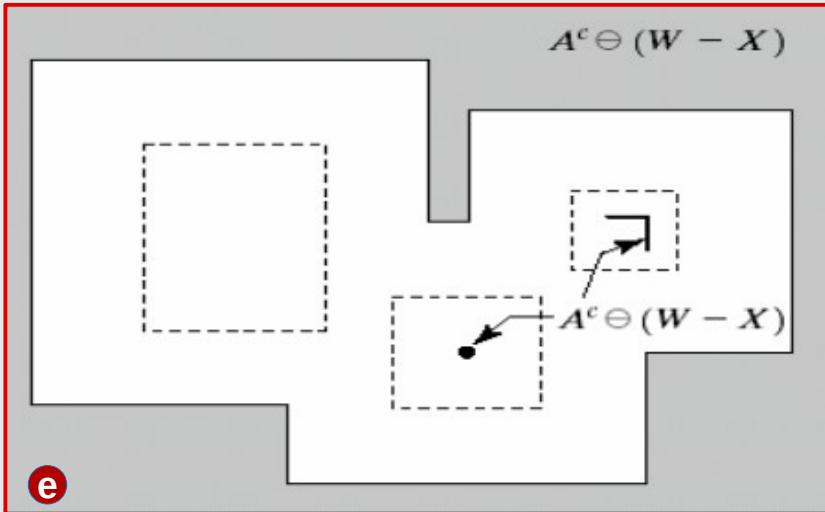
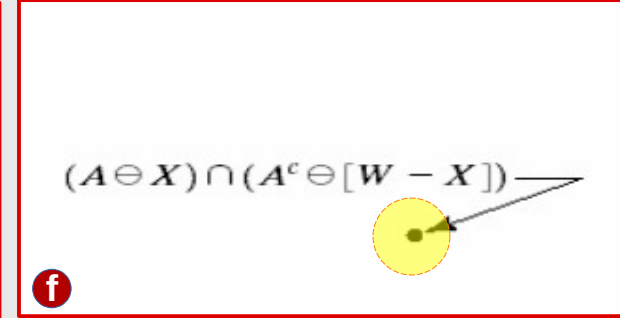
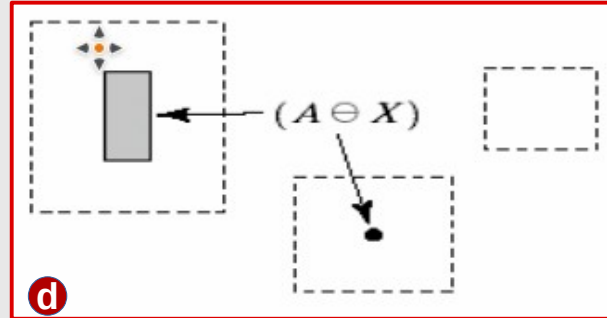
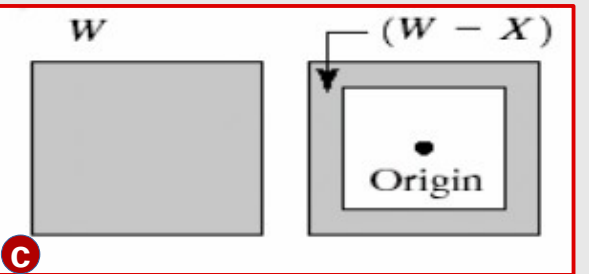
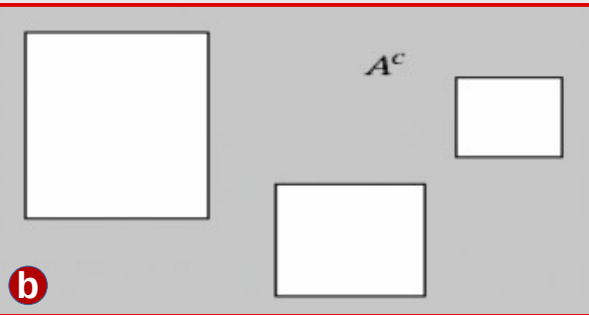
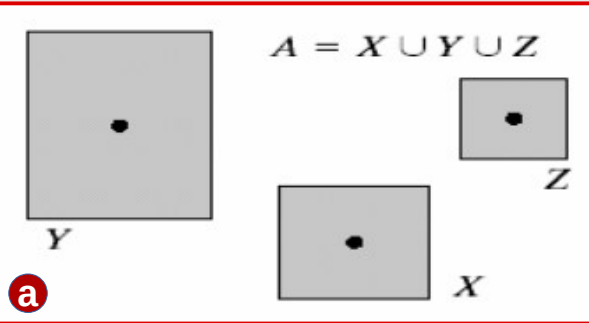
The Hit-and-Miss Transformation

(The whole process)

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$$A \otimes B = (A \ominus X) \cap [A^c \ominus (W - X)]$$

Where: $B = (X, W - X)$



- (a) Set A
- (b) Complement of A
- (c) Window W (larger than X)
- (d) Erosion of A by X
- (e) Erosion of A^c by $(W - X)$
- (f) Intersection of (d) and (e)

Search for locations of Object X (Surrounded by predefined border) within Set A

The Hit-and-Miss Transformation

- Used to look for particular patterns of foreground and background pixels
- Very simple object recognition
- All other morphological operations can be derived from it!!
- Input:
 - Binary Image
 - Structuring Element, containing 0s and 1s!!

The Hit-and-Miss Transformation

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- Example for a Hit-and-miss Structuring Element
- Contains 0s, 1s and *don't care*'s.
- Usually a “1” at the origin!

	1	
0	1	1
0	0	

The Hit-and-Miss Transformation

- Similar to Pattern Matching:
- If foreground and background pixels in the structuring element *exactly match* foreground and background pixels in the image, **then** the pixel underneath the origin of the structuring element is set to the foreground color.

Corner Detection with Hit-and-miss Transform

60

- Structuring Elements representing four corners

	1	
0	1	1
0	0	

	1	
1	1	0
	0	0

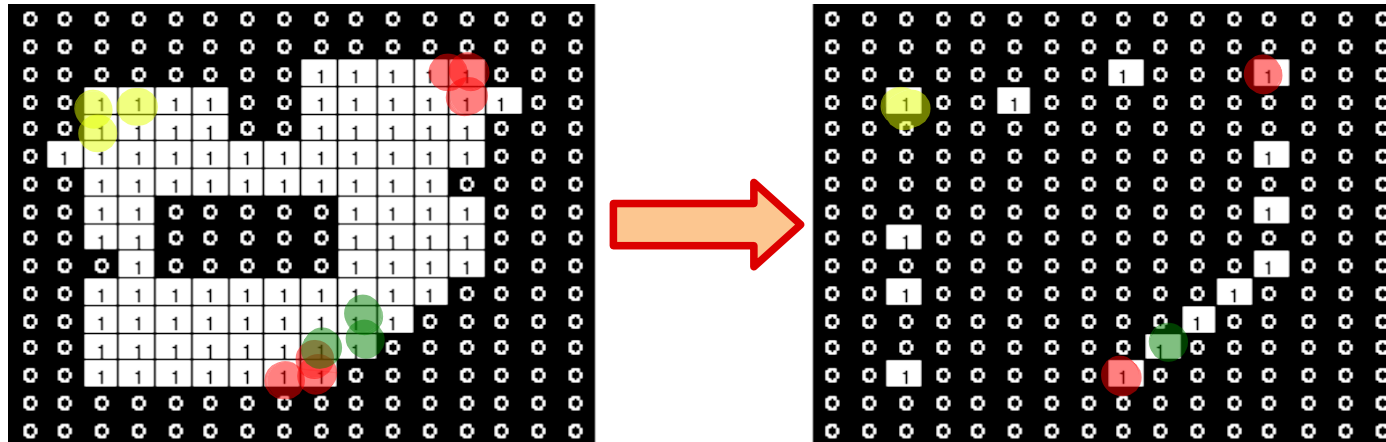
	0	0
1	1	0
	1	

0	0	
0	1	1
	1	

Corner Detection with Hit-and-miss Transform

61

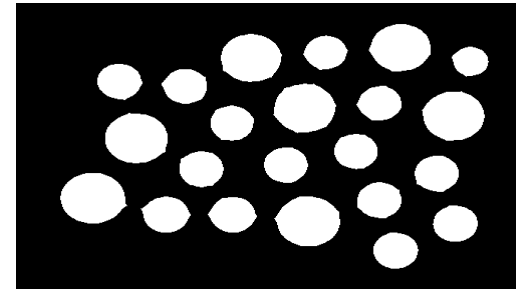
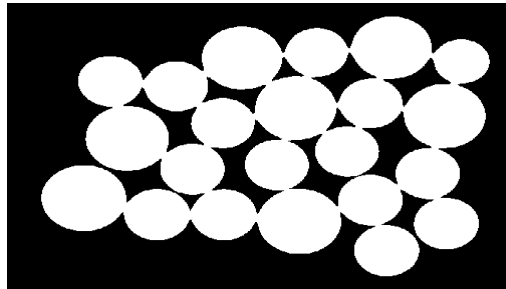
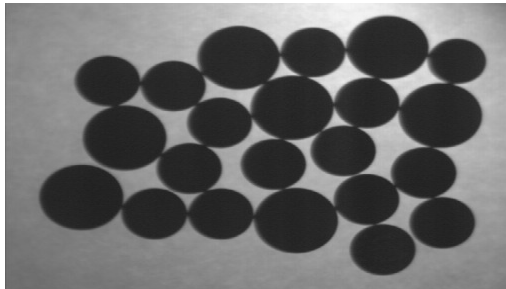
- Apply each Structuring Element
- Use OR operation to combine the four results



Applications of Image Morphology

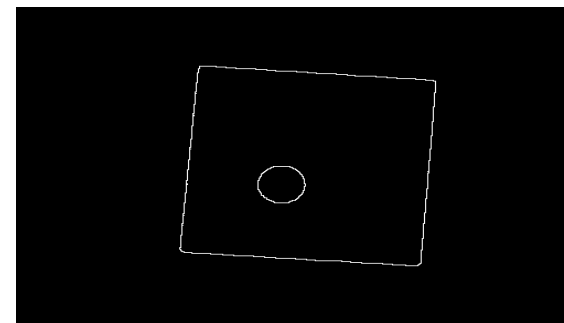
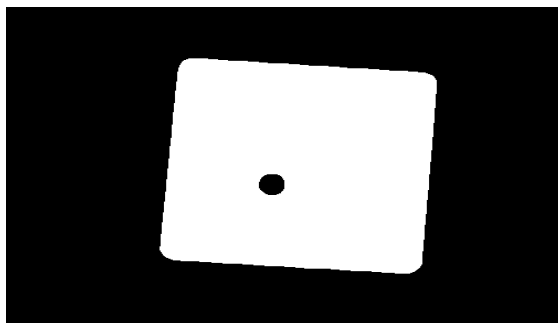
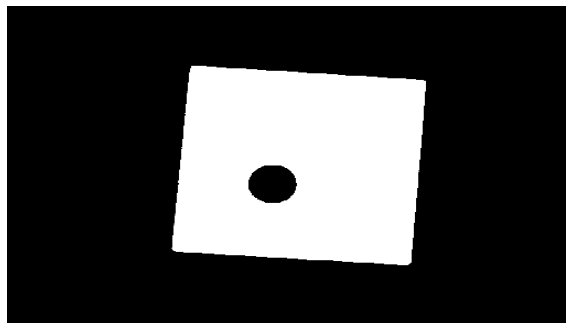
[1] Counting Coins

- Counting coins is difficult because they touch each other!
- Solution: Binarization and Erosion separates them!



[2] Edge Detection

- Edge Detection
 1. Dilate input image
 2. Subtract input image from dilated image
 3. Edges remain!

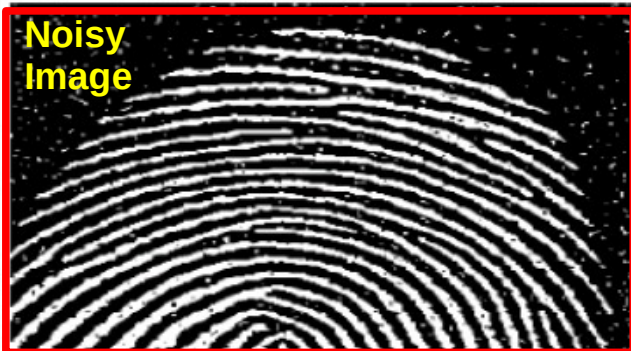


[3] Fingerprint Enhancement

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Noisy Image



A

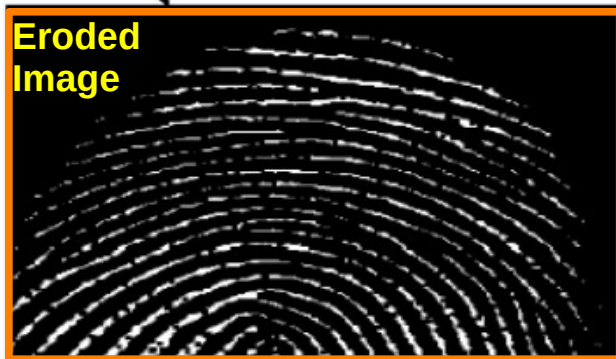
$$A \ominus B$$

$$B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

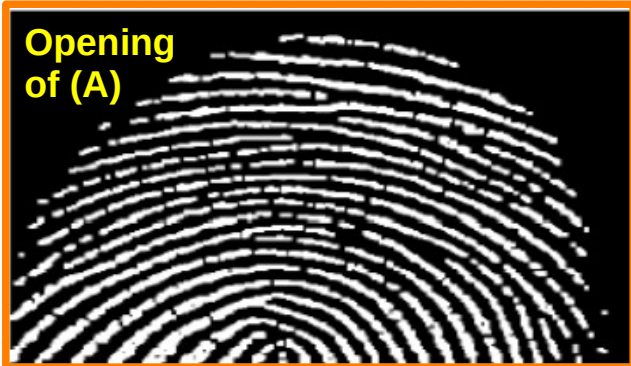
[1] Remove noise

[2] Fill Gaps

Eroded Image



Opening of (A)



$$(A \ominus B) \oplus B = A \circ B$$

$$(A \circ B) \oplus B$$

$$[(A \circ B) \oplus B] \ominus B = (A \circ B) \cdot B$$

Dilation After Opening



Closing After Dilation



original image



erosion



opening of A



1	1	1
1	1	1
1	1	1

B

dilation of the opening

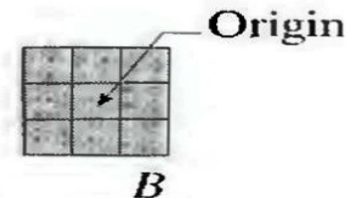
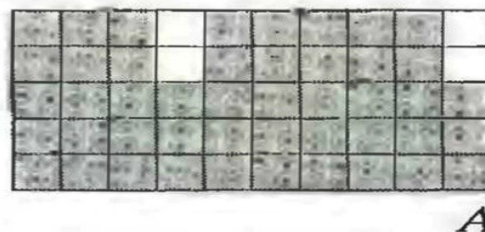


closing of the opening

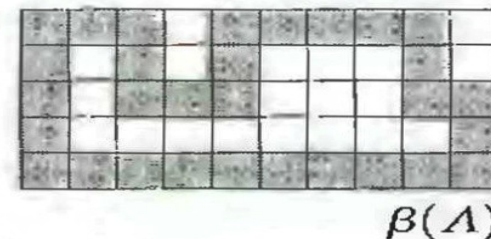
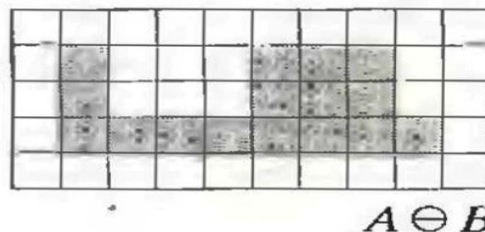


[4] Boundary Extraction

- First, **erode** A by B,
- Second, make set **difference** between A and the erosion
-
- The thickness of the contour depends on the size of constructing object – B



$$\beta(A) = A - (A \ominus B)$$



Boundary Extraction - Example



[5] Region Filling

- The following procedure can fill the region:

$$X_k = (X_{k-1} \oplus B) \cap A^c$$

Dilation = Expand = **Fill**
BUT **NOT** in Background (A^c)

Where:

$X_0 = p$ (the initial point we should assign), and

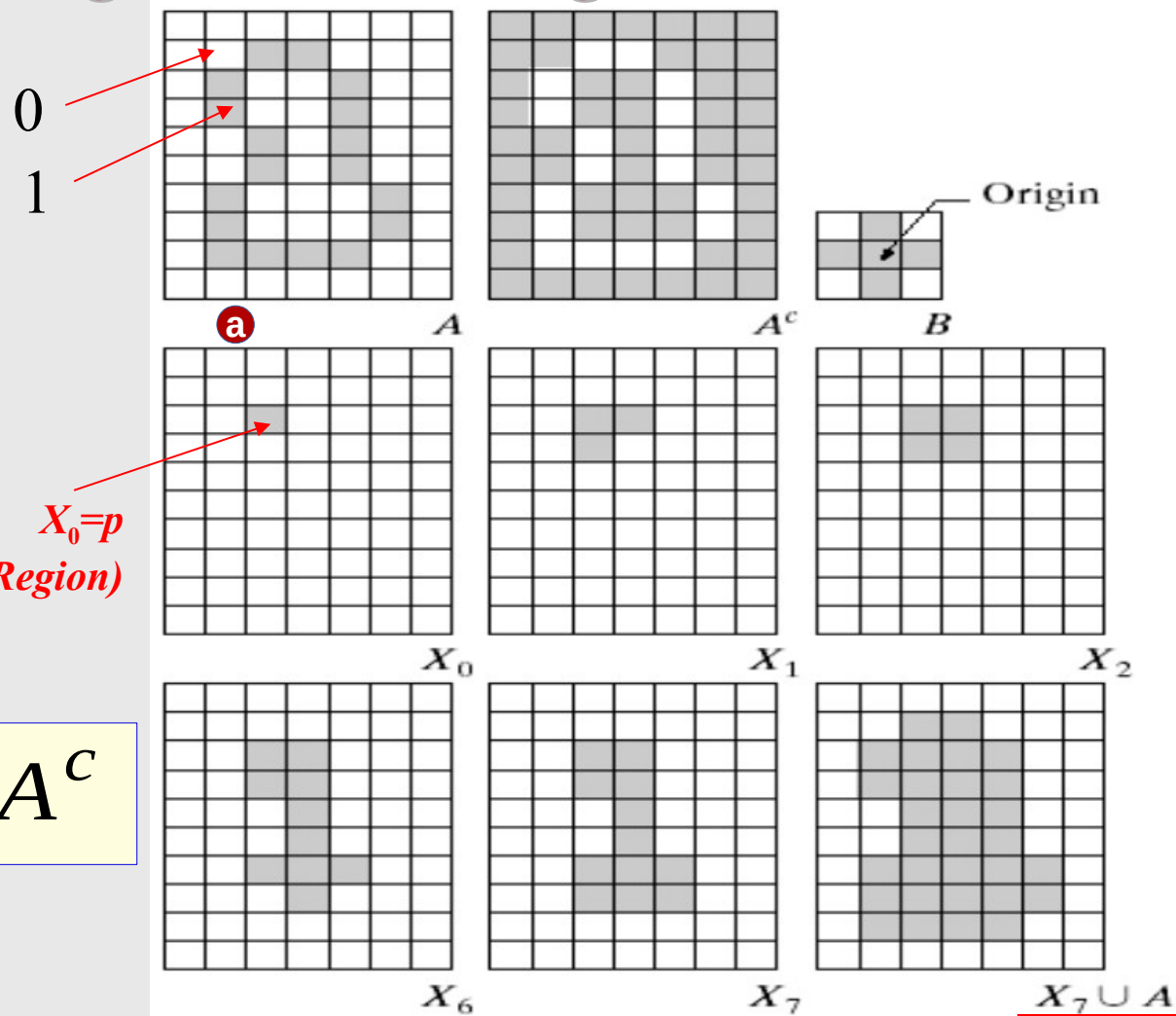
B is the symmetric structuring element

[5] Region Filling

- (a) Set **A**
- (b) Complement of **A**
- (c) Structuring Element **B**
- (d) Initial Point inside Boundary
- (*) Various filling steps
- (**) Final Result Union **A**

$X_0 = p$
(Inside Region)

$$X_k = (X_{k-1} \oplus B) \cap A^c$$

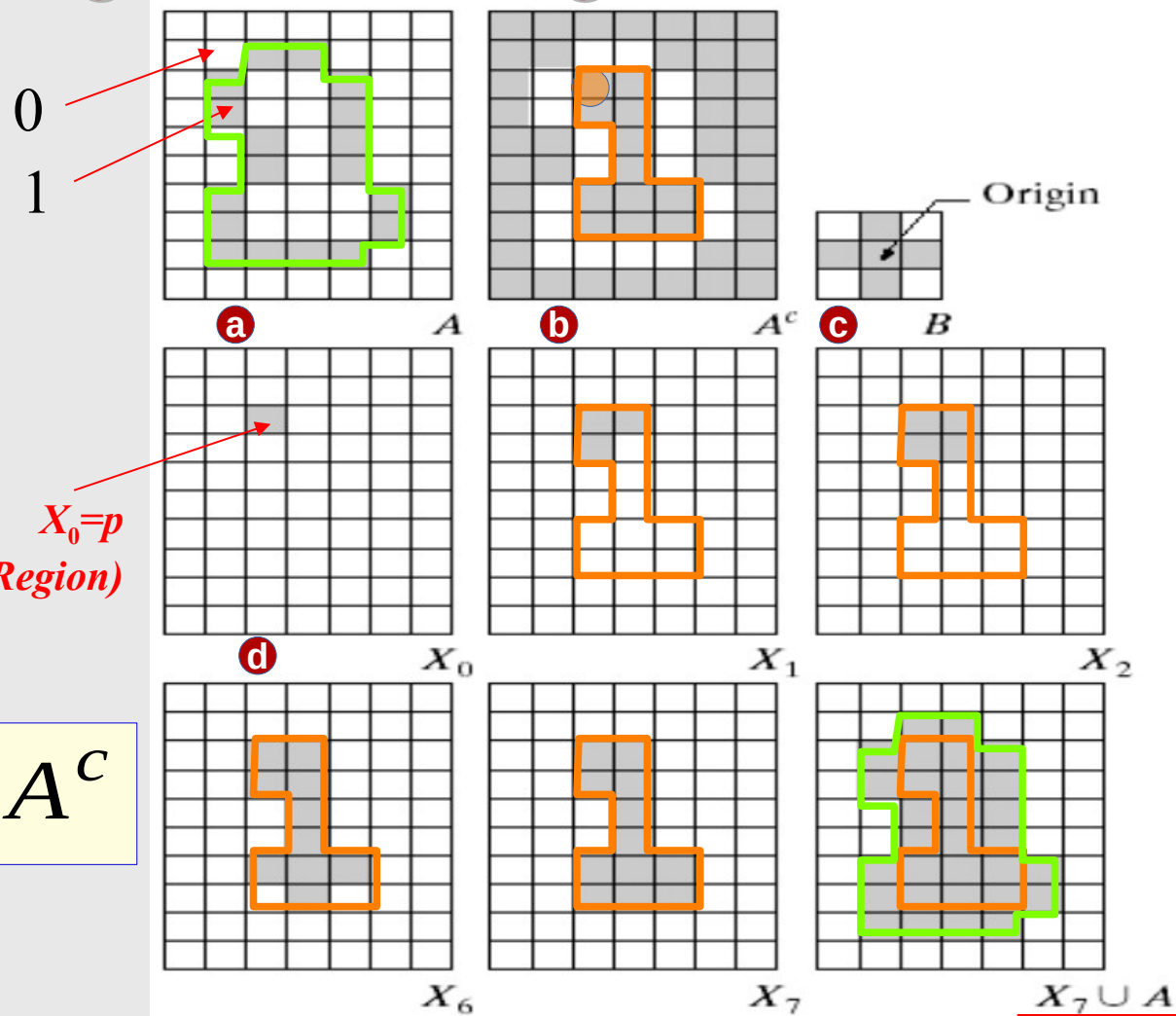


[5] Region Filling

- (a) Set **A**
- (b) Complement of **A**
- (c) Structuring Element **B**
- (d) Initial Point inside Boundary
- (*) Various filling steps
- (**) Final Result Union **A**

$X_0 = p$
(Inside Region)

$$X_k = (X_{k-1} \oplus B) \cap A^c$$



[6] Extraction of Connected Components

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$$X_k = (X_{k-1} \oplus B) \cap A$$

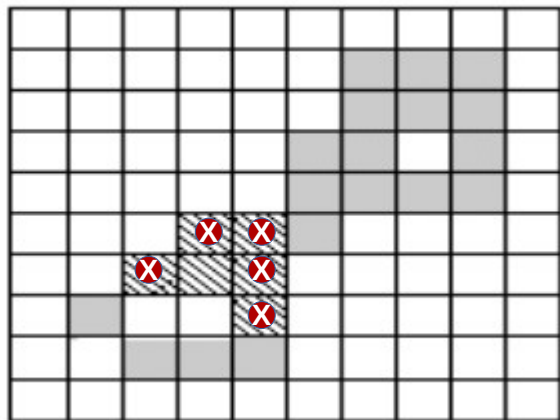
(a) Set **A** , initial point **P** (Should be “1”)

(b) Structuring Element

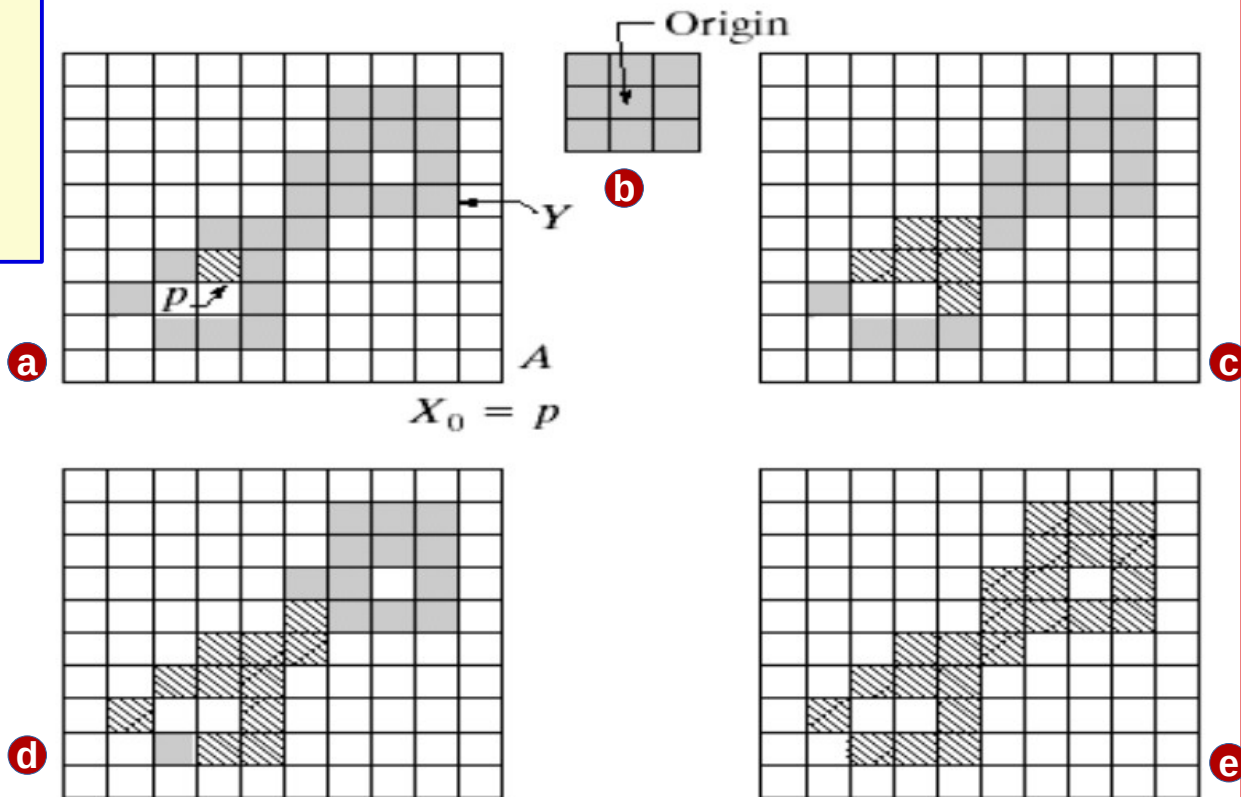
(c) Result of **first** Iteration

(d) Result of **Second** Iteration

(e) **Final** Iteration (till No Change)



From First Iteration to Second Iteration
(locations to fit Center of Structuring Element)



[7] Thinning

1. Used to **remove** selected **foreground pixels** from binary images
2. After edge detection, lines are often **thicker than one pixel**.
3. Thinning can be used to thin those line to **one pixel width**.

Thinning

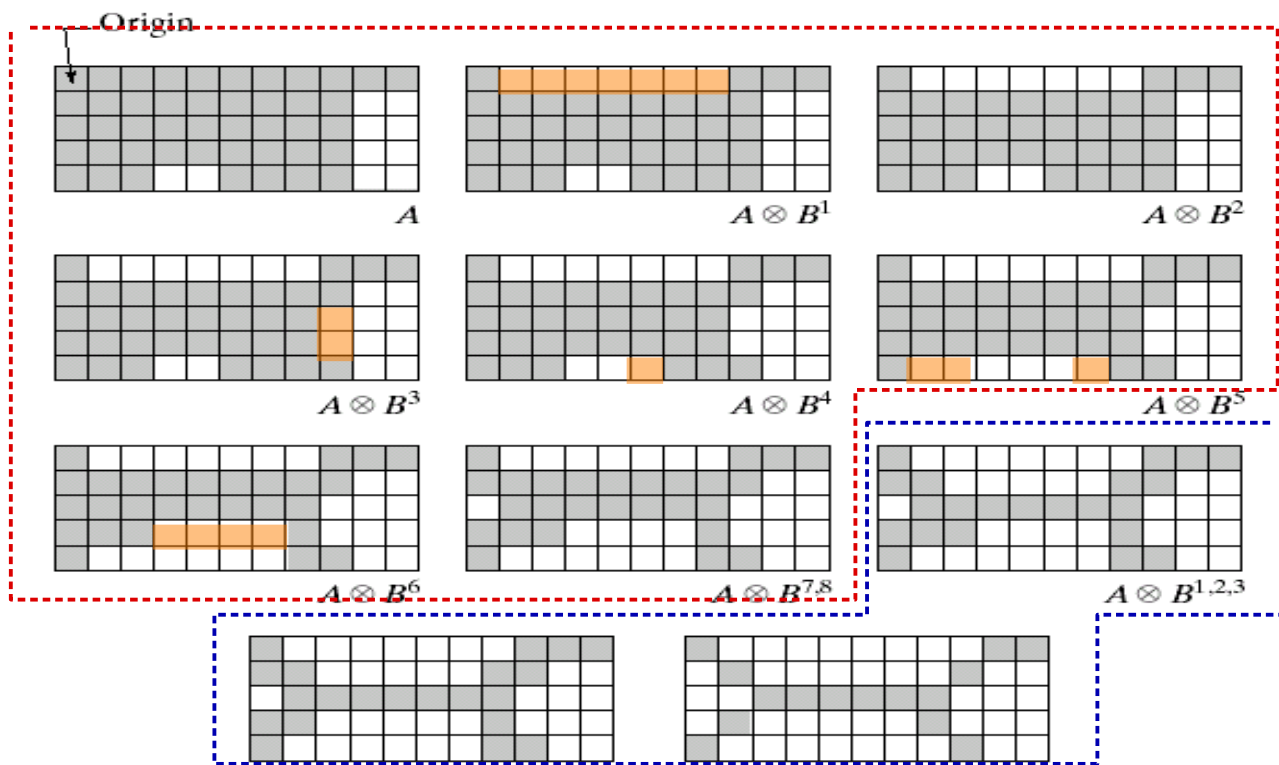
- The thinning of a set A by a structuring element B can be defined in terms of the hit-or-miss transform:

$$\begin{aligned} A \otimes B &= A - (A \circledast B) \\ &= A \cap (A \circledast B)^c \end{aligned}$$

- A more useful expression for thinning A symmetrically is based on a sequence of structuring elements:

$$\begin{aligned} \{B\} &= \{B^1, B^2, B^3, \dots, B^n\} \\ A \otimes \{B\} &= (((\dots((A \otimes B^1) \otimes B^2) \dots) \otimes B^n) \end{aligned}$$

C Corner
E Edge



- (a) Sequence of Rotated Structuring Elements B^1 B^2 B^3 ..., B^8 (for Edges and Corners)
 (b) Set "A"
 (c) Result of Apply Structuring Elements (in Sequence)
 (d) Repeat Applying All Structuring elements till no Change

Reference

		Comments
		(The Roman numerals refer to the structuring elements shown in Fig. 9.26).
Operation	Equation	
Translation	$(A)_z = \{w w = a + z, \text{ for } a \in A\}$	Translates the origin of A to point z .
Reflection	$\hat{B} = \{w w = -b, \text{ for } b \in B\}$	Reflects all elements of B about the origin of this set.
Complement	$A^c = \{w w \notin A\}$	Set of points not in A .
Difference	$A - B = \{w w \in A, w \notin B\}$ $= A \cap B^c$	Set of points that belong to A but not to B .
Dilation	$A \oplus B = \{z (\hat{B})_z \cap A \neq \emptyset\}$	"Expands" the boundary of A . (I)
Erosion	$A \ominus B = \{z (B)_z \subseteq A\}$	"Contracts" the boundary of A . (I)
Opening	$A \circ B = (A \ominus B) \oplus B$	Smooths contours, breaks narrow isthmuses, and eliminates small islands and sharp peaks. (I)
Closing	$A \bullet B = (A \oplus B) \ominus B$	Smooths contours, fuses narrow breaks and long thin gulfs, and eliminates small holes. (I)

Hit-or-miss transform	$A \circledast B = (A \ominus B_1) \cap (A^c \ominus B_2)$ $= (A \ominus B_1) - (A \oplus \hat{B}_2)$	The set of points (coordinates) at which, simultaneously, B_1 found a match ("hit") in A and B_2 found a match in A^c .
Boundary extraction	$\beta(A) = A - (A \ominus B)$	Set of points on the boundary of set A . (I)
Region filling	$X_k = (X_{k-1} \oplus B) \cap A^c; X_0 = p \text{ and } k = 1, 2, 3, \dots$	Fills a region in A , given a point p in the region. (II)
Connected components	$X_k = (X_{k-1} \oplus B) \cap A; X_0 = p \text{ and } k = 1, 2, 3, \dots$	Finds a connected component Y in A , given a point p in Y . (I)
Convex hull	$X_k^i = (X_{k-1}^i \circledast B^i) \cup A; i = 1, 2, 3, 4;$ $k = 1, 2, 3, \dots; X_0^i = A; \text{ and}$ $D^i = X_{\text{conv}}^i.$	Finds the convex hull $C(A)$ of set A , where "conv" indicates convergence in the sense that $X_k^i = X_{k-1}^i$. (III)

Comments

(The Roman numerals refer to the structuring elements shown in Fig. 9.26).

Operation

Equation

Thinning

$$\begin{aligned}
 A \otimes B &= A - (A \circledast B) \\
 &= A \cap (A \circledast B)^c \\
 \\
 A \otimes \{B\} &= \\
 ((\dots((A \otimes B^1) \otimes B^2) \dots) \otimes B^n) \\
 \{B\} &= \{B^1, B^2, B^3, \dots, B^n\}
 \end{aligned}$$

Thins set A . The first two equations give the basic definition of thinning. The last two equations denote thinning by a sequence of structuring elements. This method is normally used in practice. (IV)

Thickening

$$\begin{aligned}
 A \odot B &= A \cup (A \circledast B) \\
 \\
 A \odot \{B\} &= \\
 ((\dots(A \odot B^1) \odot B^2 \dots) \odot B^n)
 \end{aligned}$$

Thickens set A . (See preceding comments on sequences of structuring elements.) Uses IV with 0's and 1's reversed.

Skeletons

$$S(A) = \bigcup_{k=0}^K S_k(A)$$

$$S_k(A) = \bigcup_{k=0}^K \{ (A \ominus kB) - [(A \ominus kB) \circ B] \}$$

Reconstruction of A :

$$A = \bigcup_{k=0}^K (S_k(A) \oplus kB)$$

Finds the skeleton $S(A)$ of set A . The last equation indicates that A can be reconstructed from its skeleton subsets $S_k(A)$. In all three equations, K is the value of the iterative step after which the set A erodes to the empty set. The notation $(A \ominus kB)$ denotes the k th iteration of successive erosion of A by B . (I)

Pruning

$$X_1 = A \otimes \{B\}$$

$$X_2 = \bigcup_{k=1}^8 (X_1 \circledast B^k)$$

$$X_3 = (X_2 \oplus H) \cap A$$

$$X_4 = X_1 \cup X_3$$

X_4 is the result of pruning set A . The number of times that the first equation is applied to obtain X_1 must be specified. Structuring elements V are used for the first two equations. In the third equation H denotes structuring element I .